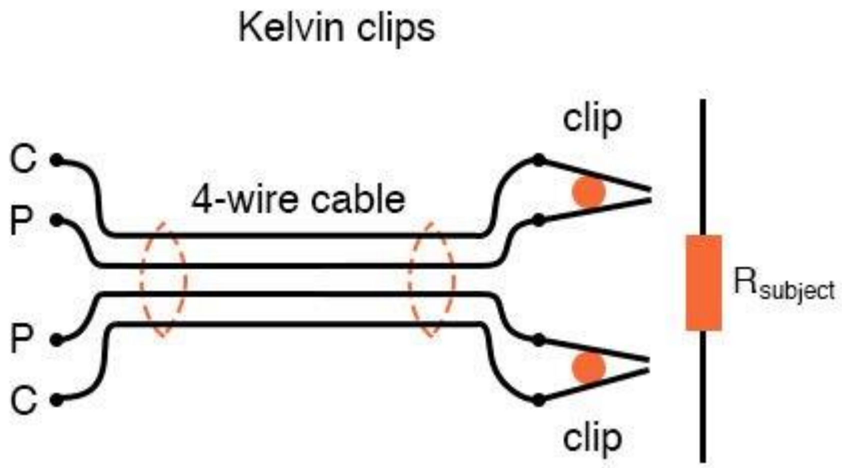
درس ابزار دقیق

اندازه گیری مقاومت



نوید علایی شینی

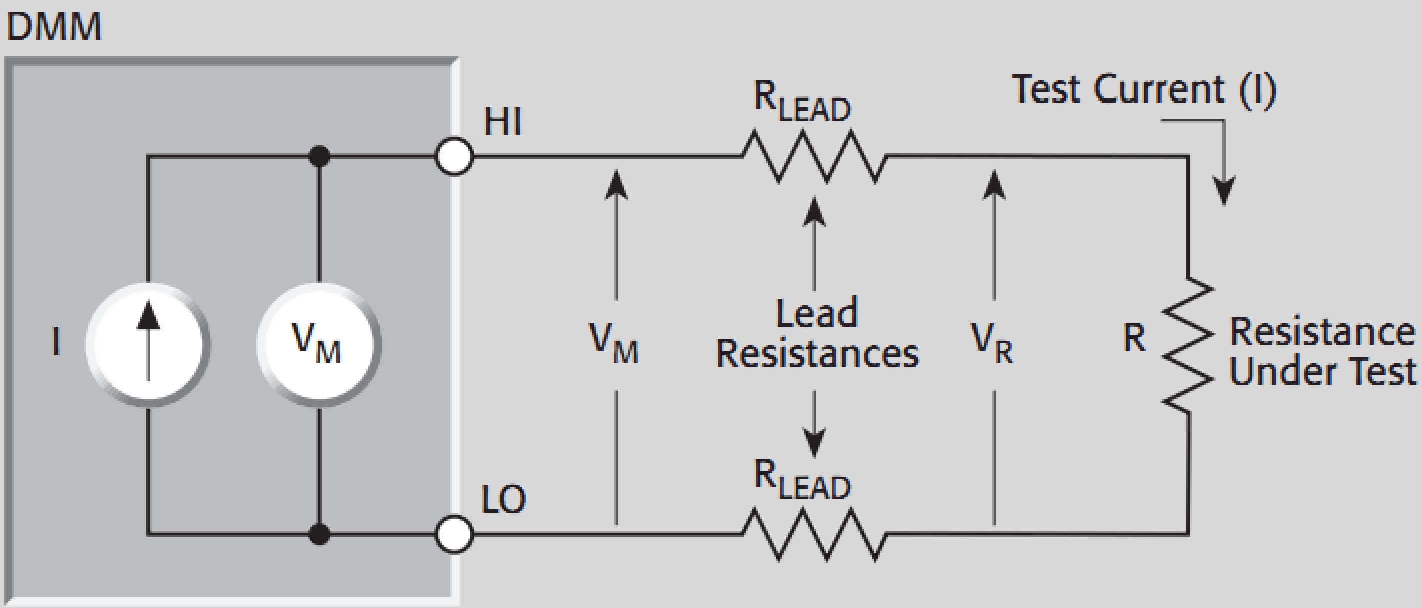
استادیار دانشگاه شهید چمران اهواز

اردیبهشت ماه ۱۴۰۳

فهرست

اندازه گیری دو سیمه

FIGURE 3-14: Two-Wire Resistance Measurement



V_M = Voltage measured by meter

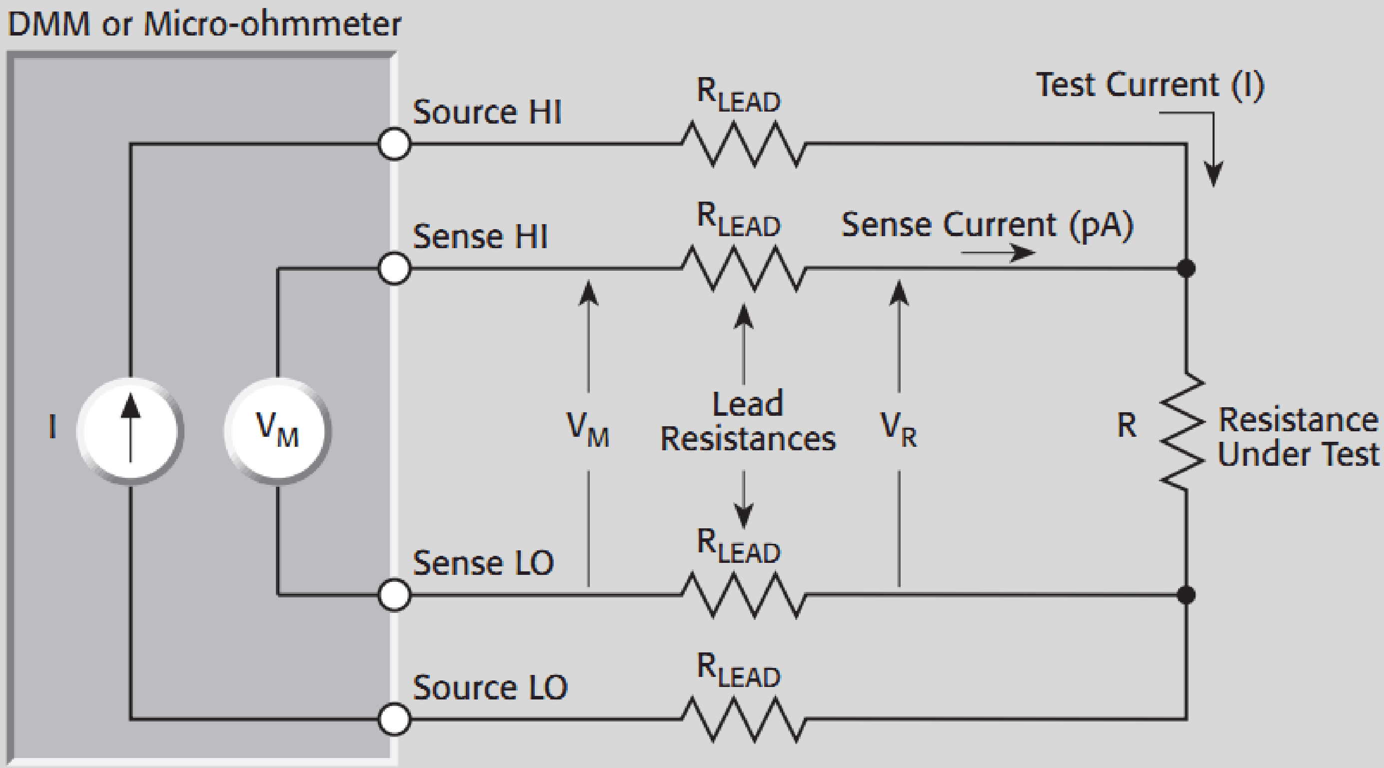
V_R = Voltage across resistor

$$\text{Measured Resistance} = \frac{V_M}{I} = R + (2 \times R_{LEAD})$$

Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

اندازه گیری چهارسیمه

FIGURE 3-15: Four-Wire Resistance Measurement



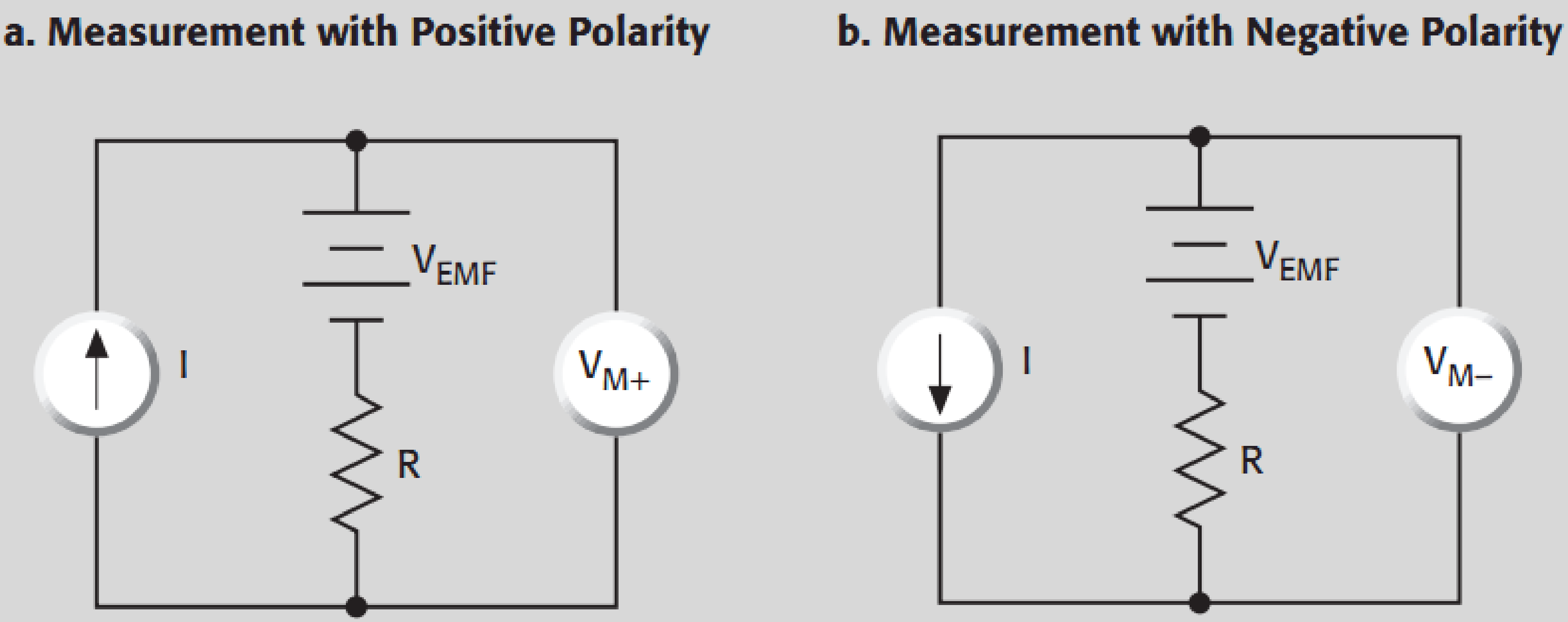
V_M = Voltage measured by meter
V_R = Voltage across resistor (R)
Because sense current is negligible, $V_M = V_R$

and measured resistance $= \frac{V_M}{I} = \frac{V_R}{I}$

Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

حذف خطای احتمالی ناشی از پدیده تولید ولتاژ حرارتی ترموالکتریک

FIGURE 3-16: Canceling Thermoelectric EMFs with Current Reversal



$$V_{M+} = V_{EMF} + I\,R$$

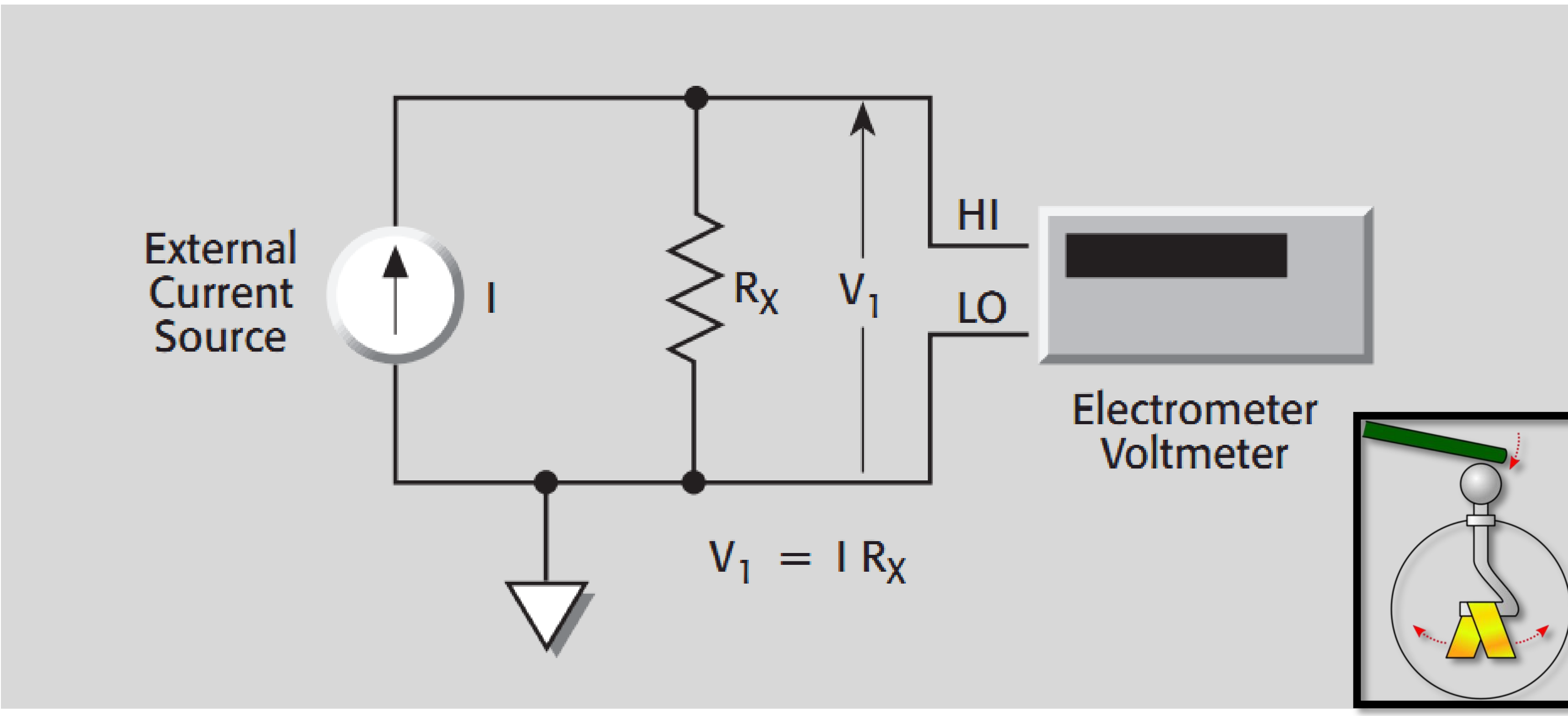
$$V_{M-} = V_{EMF} - I\,R$$

$$V_M = \frac{V_{M+} - V_{M-}}{2} = I\,R$$

Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

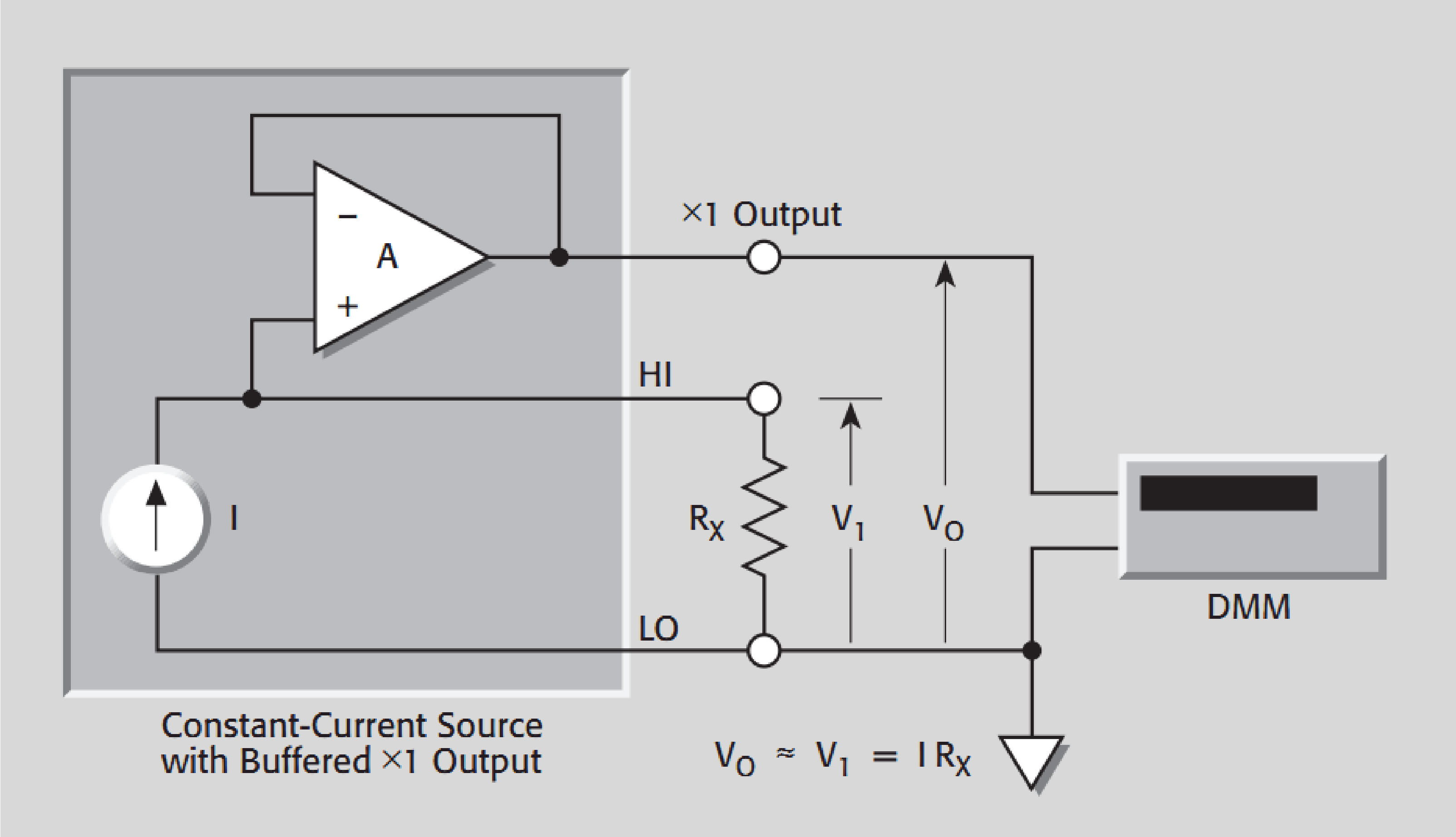
اندازه گیری مقاومتهای بزرگ

FIGURE 1-20: High Resistance Measurement Using External Current Source with Electrometer Voltmeter



- **A modern electrometer is a highly sensitive electronic voltmeter** whose input impedance is so high that the current flowing into it can be considered, for most practical purposes, to be zero.

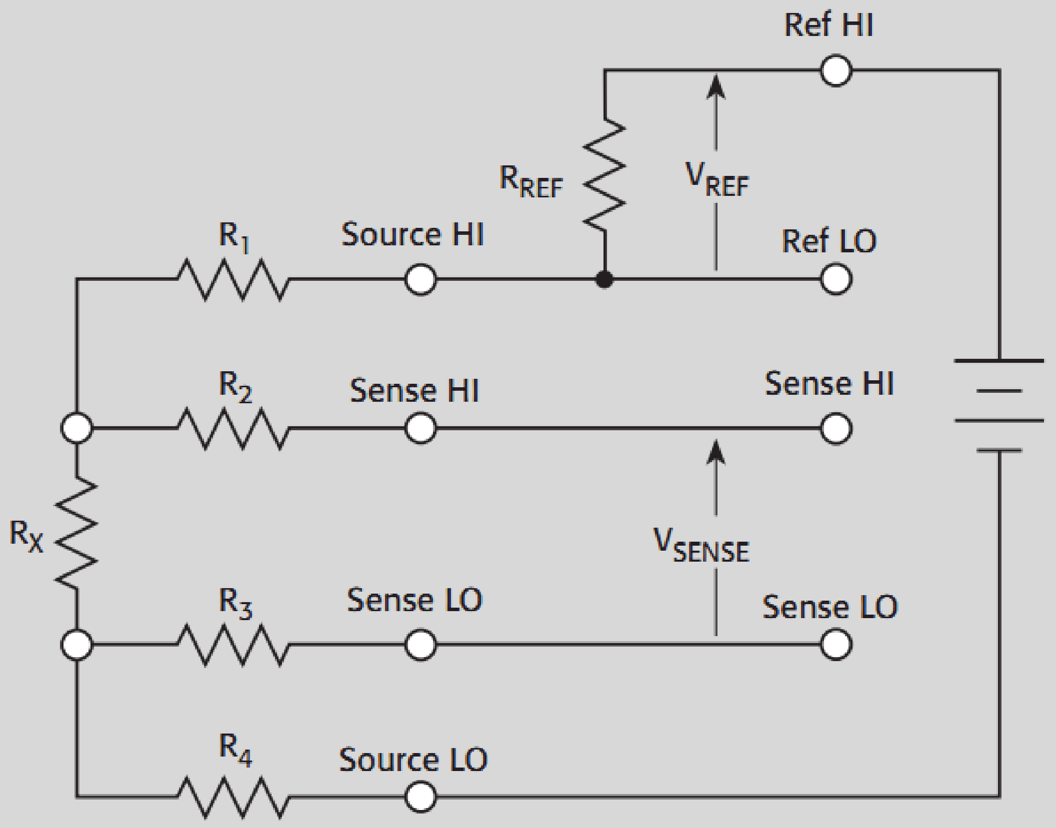
FIGURE 1-21: High Resistance Measurement Using a True Current Source with a DMM



Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

اندازه گیری مقاومتهای کوچک

FIGURE 1-23: Micro-ohmmeter Resistance Measurement



$$R_X = R_{REF} \cdot \frac{V_{SENSE}}{V_{REF}}$$

Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

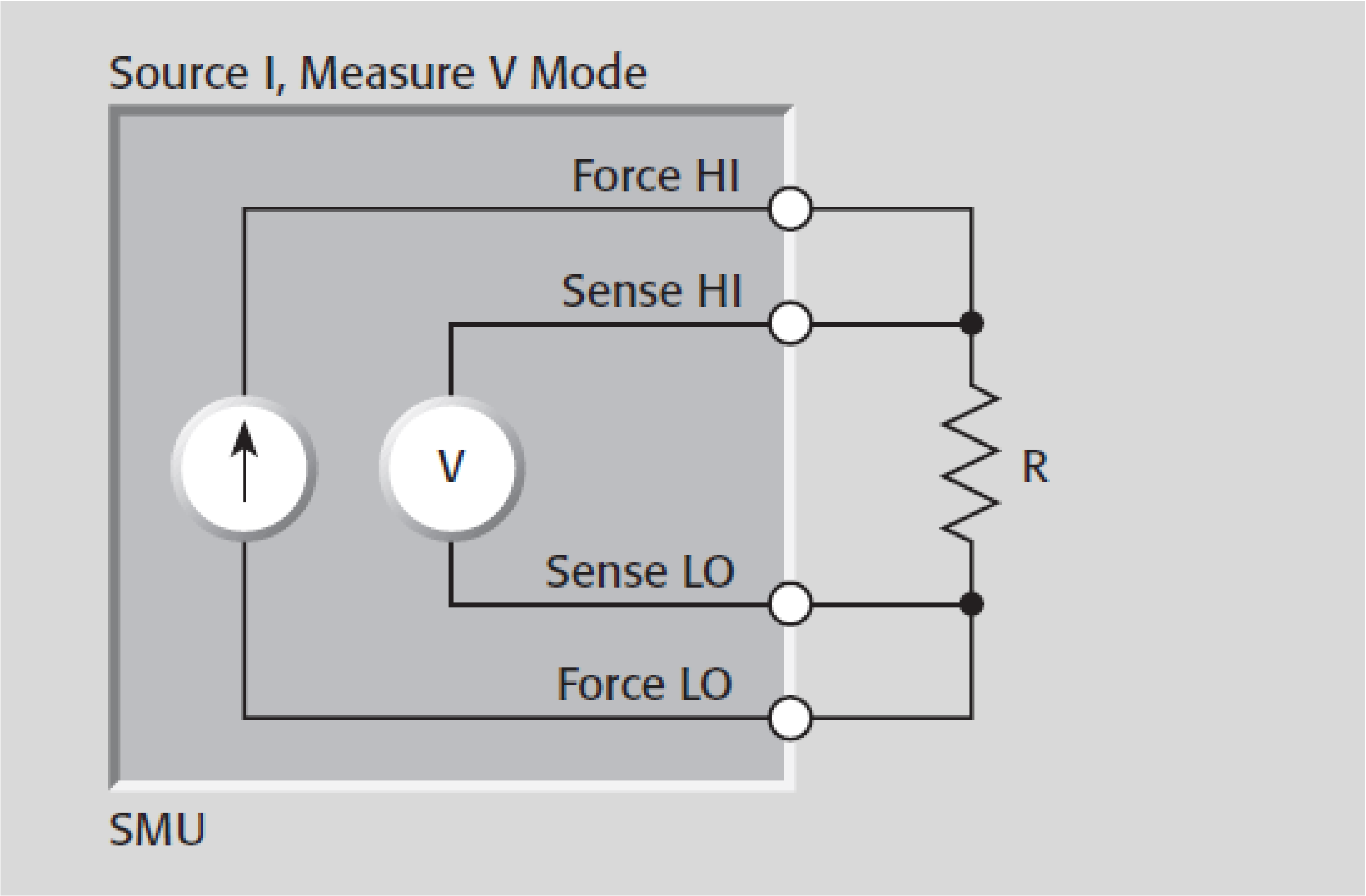
The Source-Measure Unit

As its name implies, a source-measure unit (SMU) has both measuring and sourcing capabilities. Adding current and voltage sourcing capabilities to a measuring instrument provides an extra degree of versatility for many low level measurement applications. For example, very high resistance values can be determined by applying a voltage across a device and measuring the resulting current. The added sourcing functions also make a SMU more convenient and versatile than using separate instruments for such applications as generating I-V curves of semiconductors and other types of devices.

SMUs have a number of electrometer-like characteristics that make them suitable for low level measurements. The input resistance is very high (typically 100TΩ or more), minimizing circuit loading when making voltage measurements from high impedance sources. The current measurement sensitivity is also similar to that of the electrometer picoammeter—typically as low as 10fA.

Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

Using the SMU in the Four-Wire Mode to Measure High Resistance



Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

Source-Measure Unit

معرفی دستگاه SMU:

The typical SMU provides the following four functions:

- Measure voltage
- Measure current
- Source voltage
- Source current

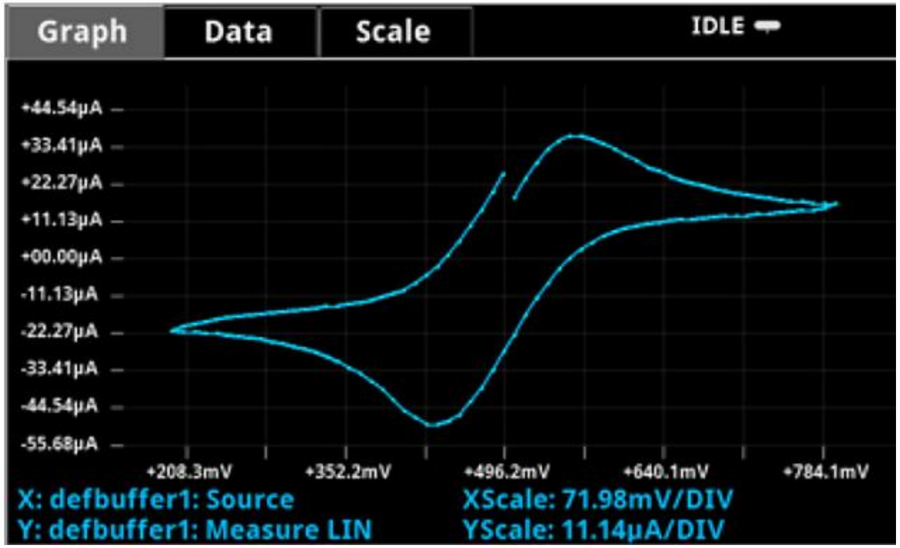
These functions can be used separately or they can be used together in the following combinations:

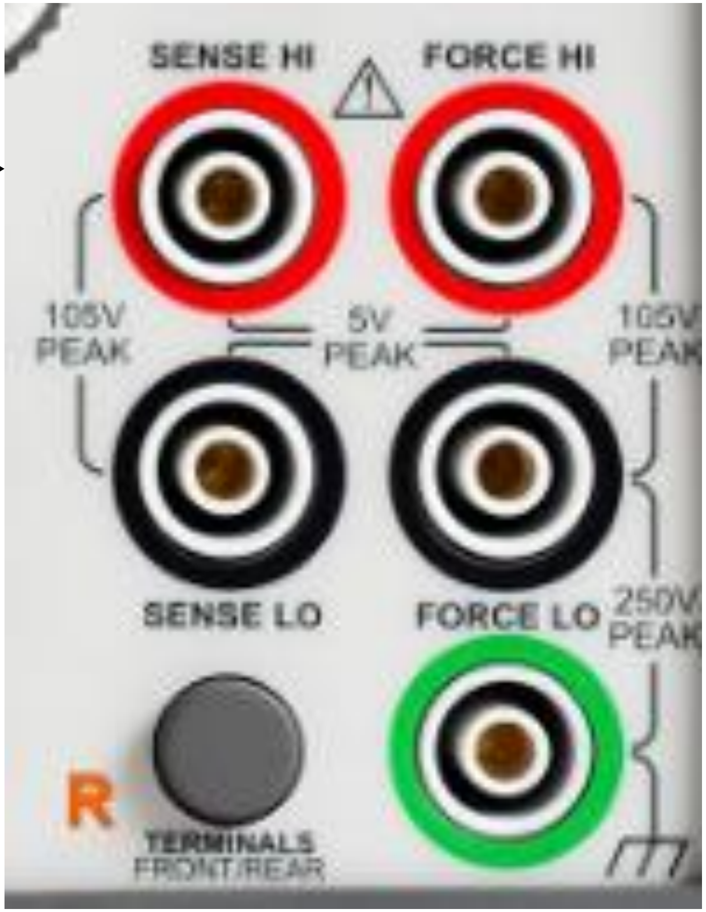
- Simultaneously source voltage and measure current, or
- Simultaneously source current and measure voltage.

نمونه دستگاه SMU:

2460 - Source Measure Unit, SourceMeter, 200mV to 100V, 1µA to 7A, 100 W

£7,060.00

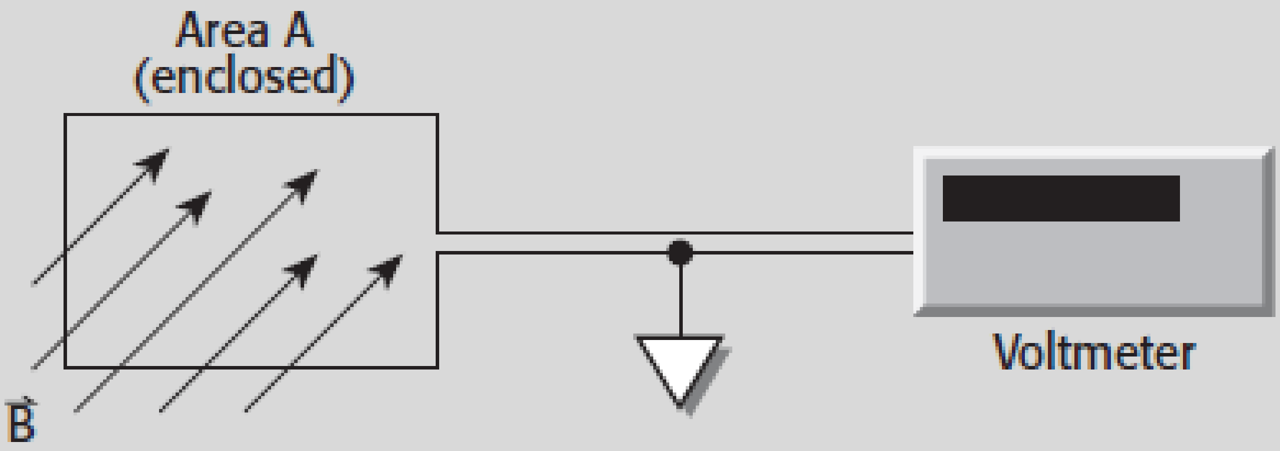




https://uk.farnell.com/keithley/2460/source-meter-instrument-100v-7a/dp/2449573

نویزگیری در اندازه گیری- القای ولتاژ بر اثر میدانهای مغناطیسی:

FIGURE 3-9: Low Voltages Generated by Magnetic Fields

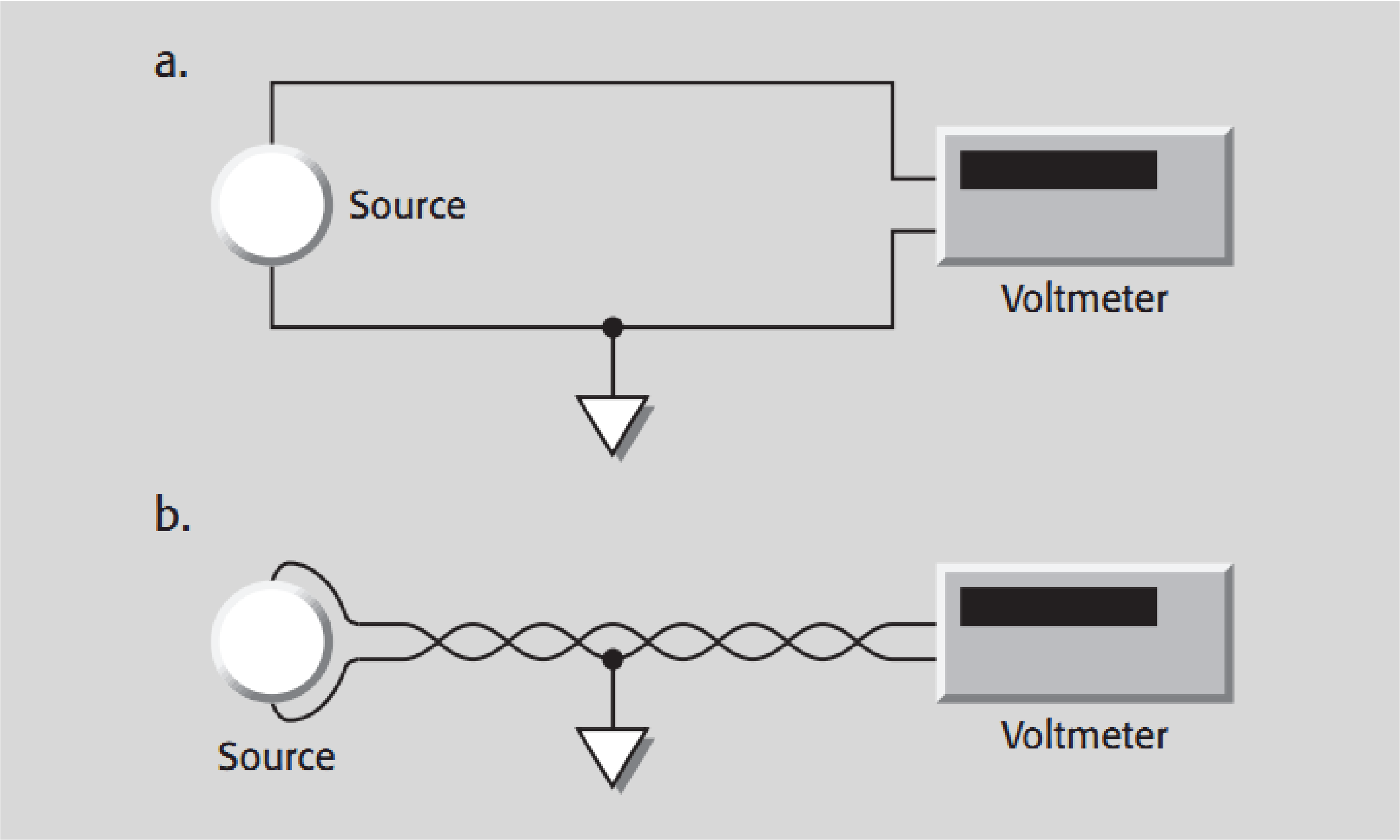


The voltage developed due to a field passing through a circuit enclosing a prescribed area is:

$$V_B = \frac{d\phi}{dt} = \frac{d(\vec{B}A)}{dt} = \vec{B}\frac{dA}{dt} + A\frac{d\vec{B}}{dt}$$

نویزگیری- کاهش اثر میدانهای مغناطیسی با استفاده از زوج سیم به هم پیچیده:

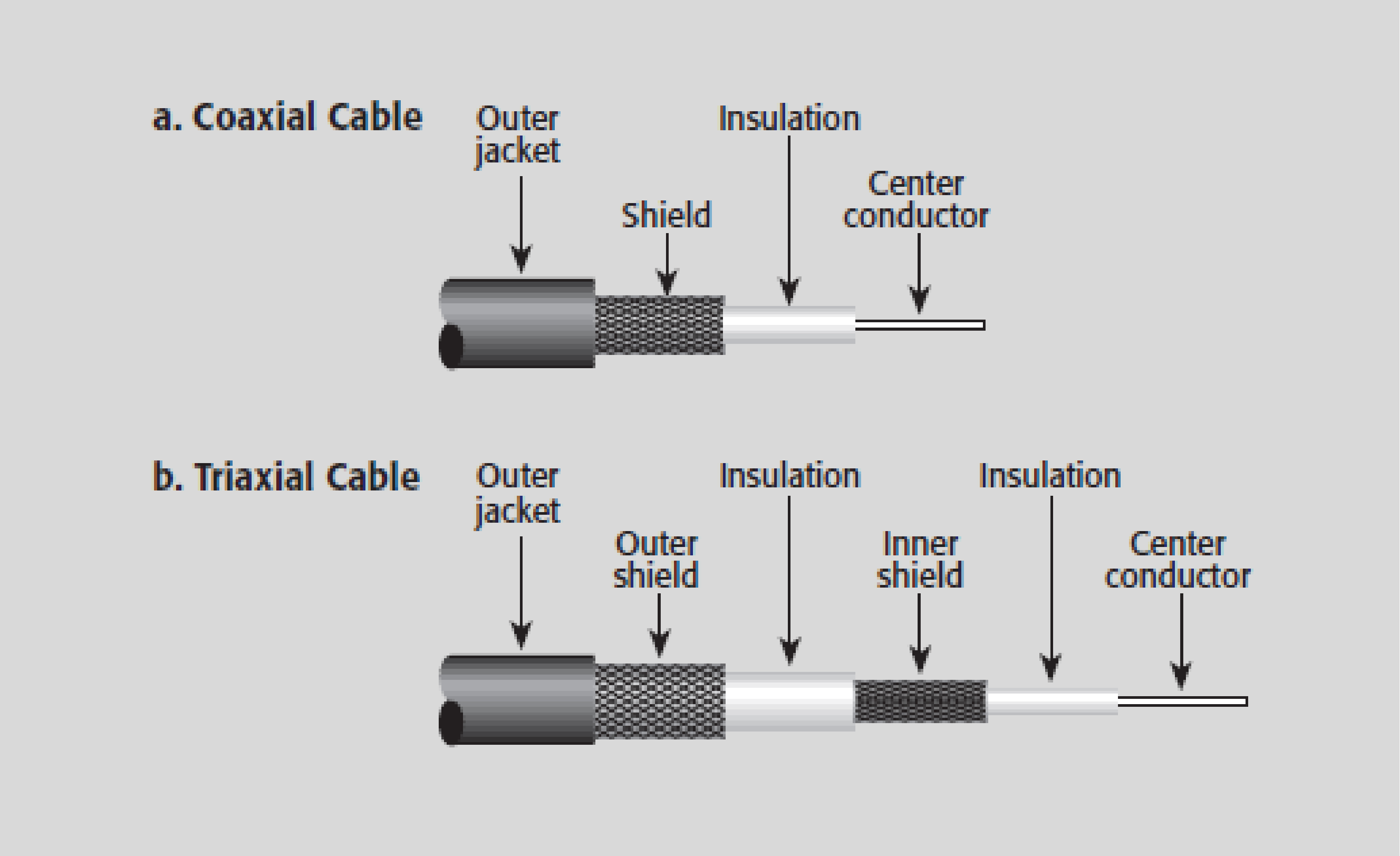
FIGURE 3-10: Minimizing Interference from Magnetic Fields



Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

نویزگیری- نمونه ای از کابلهای دارای محافظ

FIGURE 2-54: Coaxial and Triaxial Cables



Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

نویزگیری- اتصال به کابلهای محافظ دار

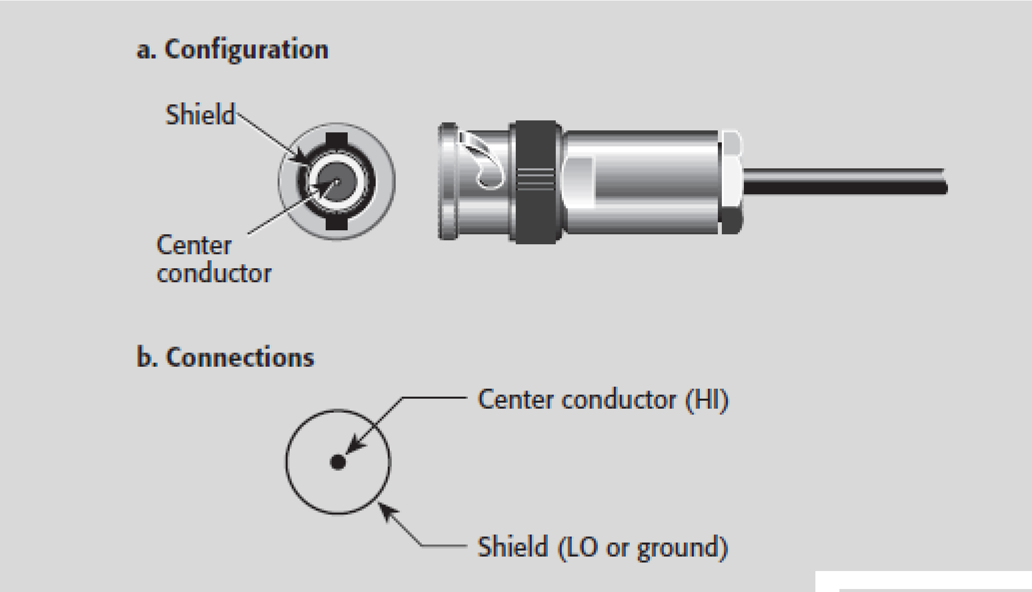


FIGURE 2-55: BNC Connector

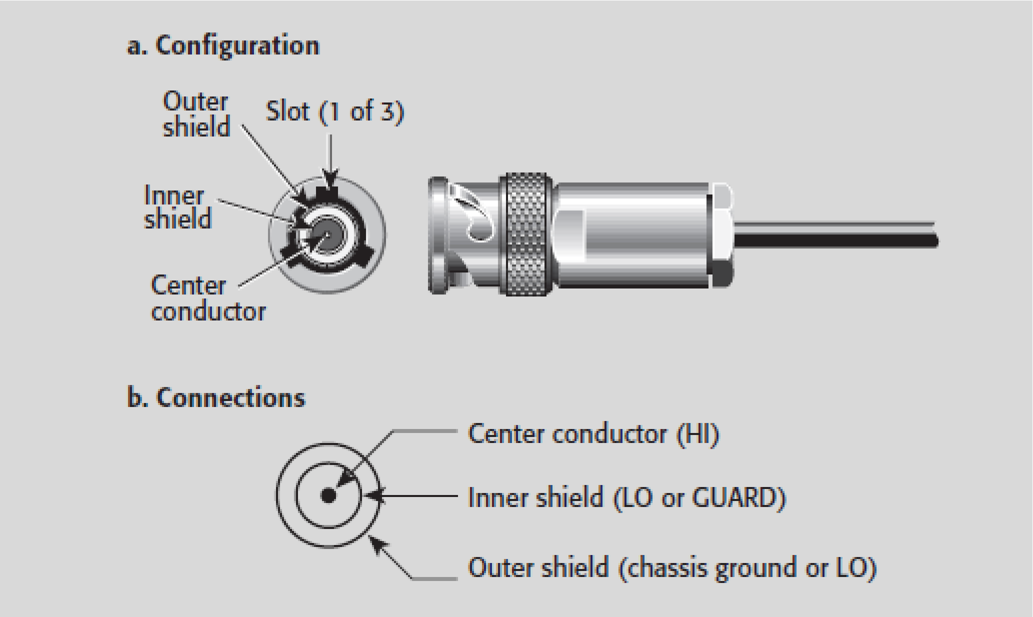
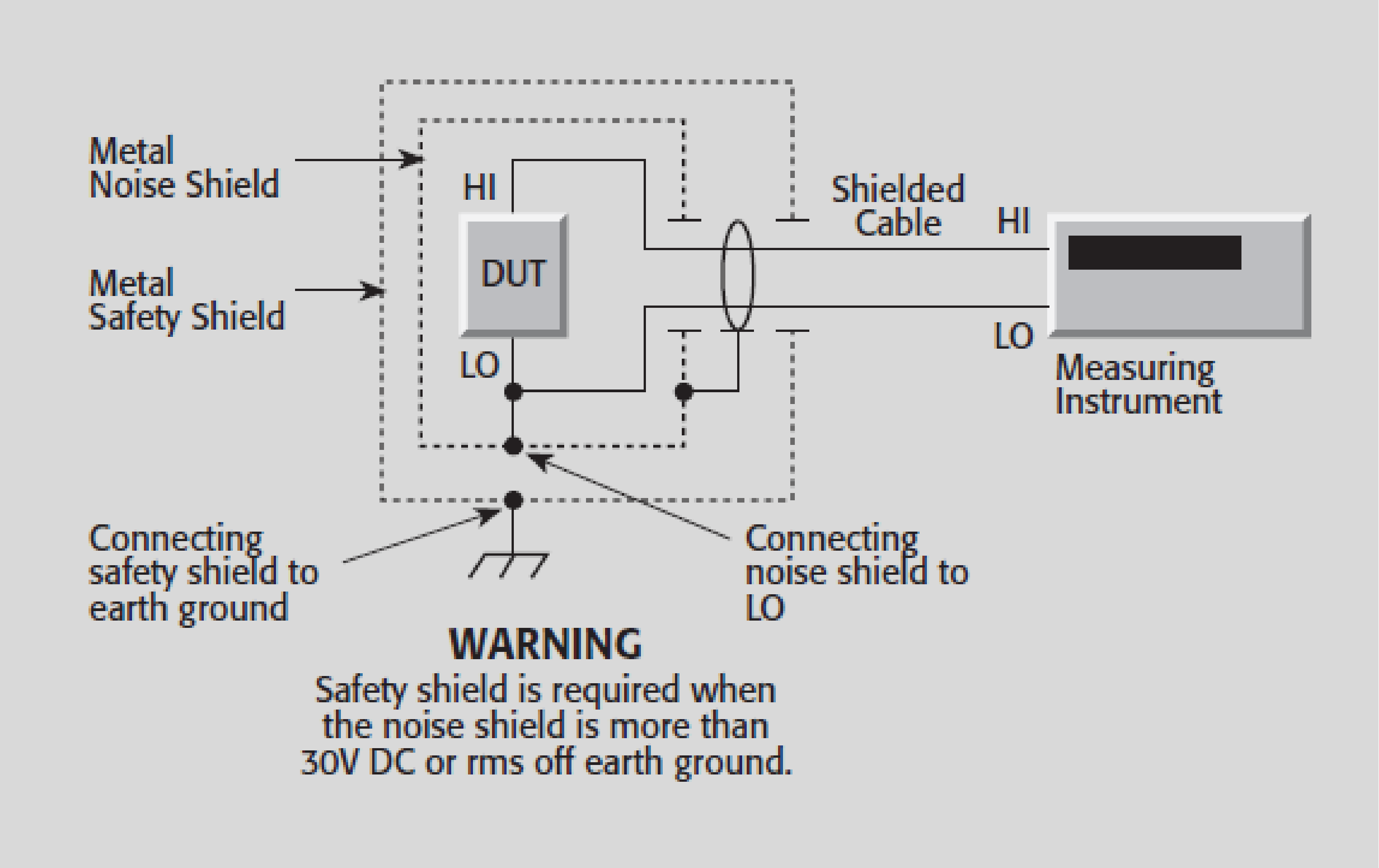


FIGURE 2-56: Three-Slot Triaxial Connector

نویزگیری- حفاظت در برابر نویز با کاهش تداخلات رادیویی/الکترومغناطیسی:

FIGURE 3-6: Shielding to Attenuate RFI/EMI Interference



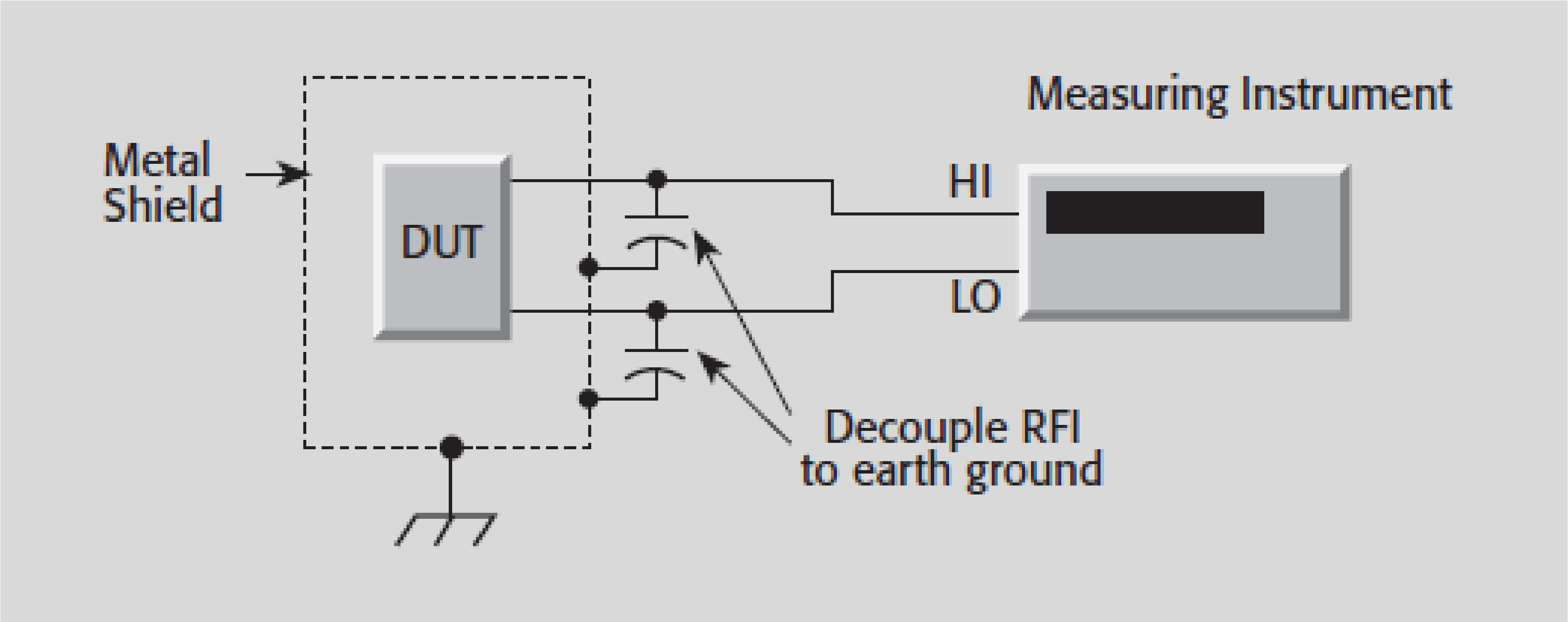
DUT: Device Under Test قطعه مورد آزمایش

Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

نویزگیری در اندازه گیری:

اتصال خازنی به محافظ در برابر نویز برای کاهش تداخلات رادیویی و الکترومغناطیسی

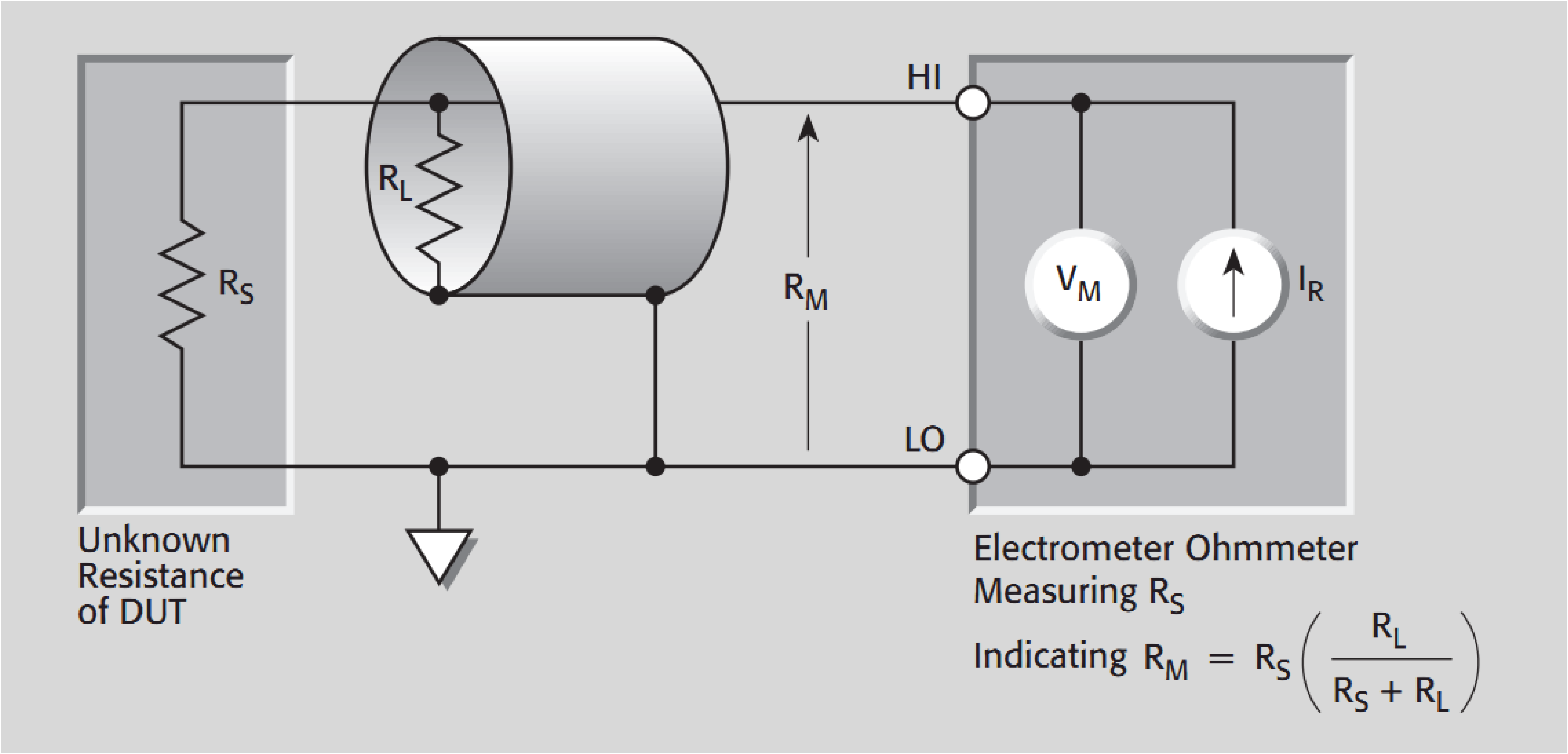
FIGURE 3-7: Shielded Connections to Reduce Inducted RFI/EMI



Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

خطا در اندازه گیری مقاومتهای بزرگ با کابل محافظ دار:

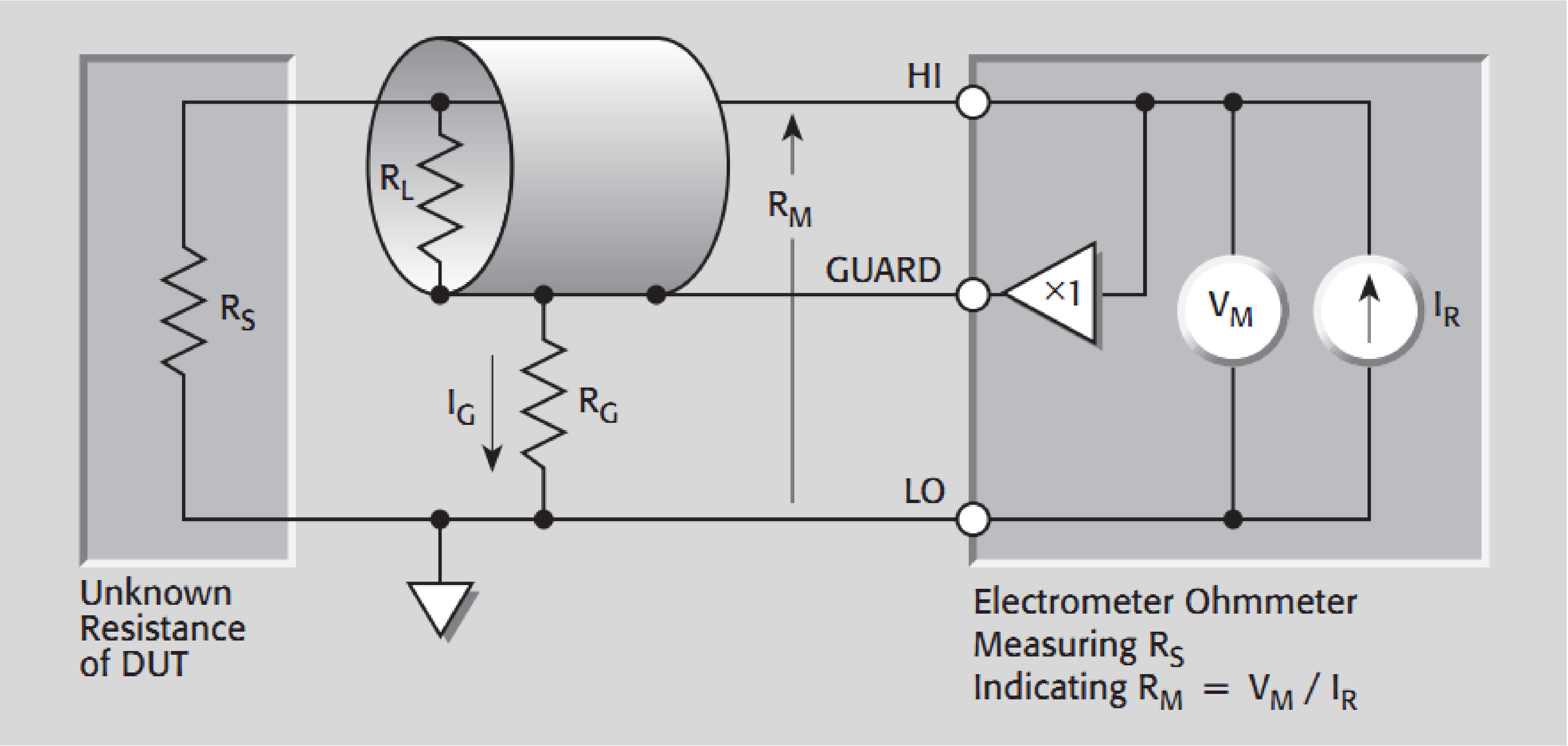
FIGURE 2-34a: Effects of Cable Resistance on High Resistance Measurements



Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

خطا در اندازه گیری مقاومتهای بزرگ در اندازه گیری با کابل محافظ دار- راه حل:

FIGURE 2-34c: Guarding Cable Shield to Eliminate Leakage Resistance



Handbook, Low Level Measurements. "*Precision DC Current, Voltage, and Resistance Measurements*." Keithley Instruments (2004).

اندازه گیری بوسیله پل وتستون

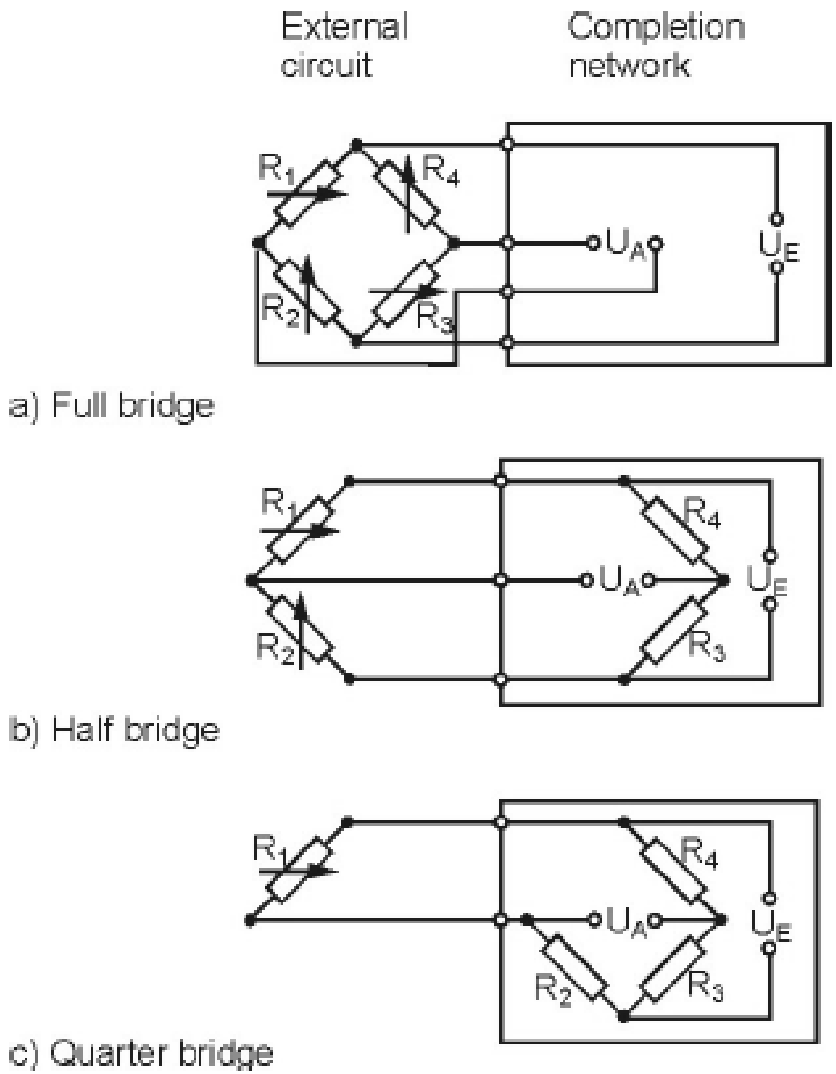
In 1843 the English physicist, Sir Charles Wheatstone (1802-1875), found a bridge circuit for measuring electrical resistances. In this bridge circuit, known today as the Wheatstone bridge circuit, unknown resistances are compared with well-defined resistances [1]. The Wheatstone bridge is also well suited for the measurement of small changes of a resistance and is, therefore, also suitable for measuring the resistance change in a strain gage (SG). It is commonly known that the strain gage transforms strain applied into a proportional change of resistance. The relationship between the applied strain ε ($\varepsilon = \Delta L/L_0$) and the relative change of the resistance of a strain gage is described by the equation

$$\frac{\Delta R}{R_0} = k \cdot \varepsilon \qquad (1)$$

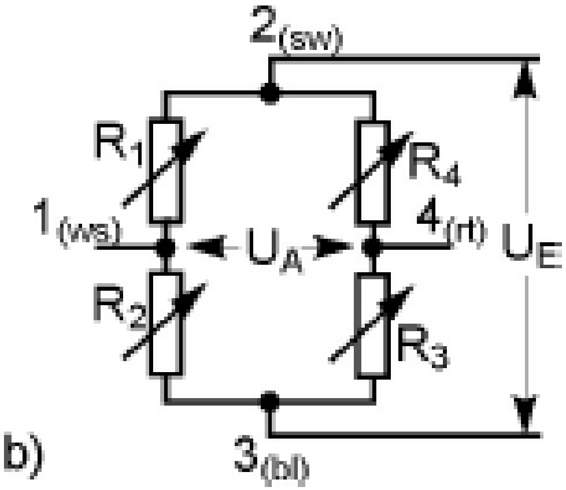
The factor k, also known as the gage factor, is a characteristic of the strain gage and has been checked experimentally [2]. The exact value is specified on each strain gage package. In general, the gage factor for metal strain gages is about 2.

Book: Hoffmann, Karl. *Applying the Wheatstone bridge circuit*. Germany: HBM, 1974.

انواع پل وتستون



اندازه گیری بوسیله پل وتستون

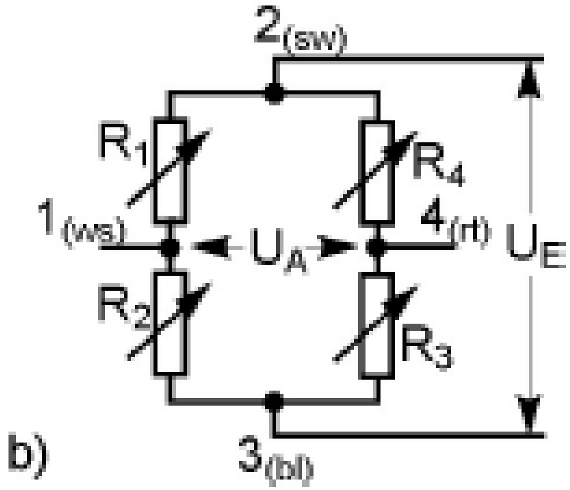


$$\frac{U_A}{U_E}=\frac{R_1}{R_1+R_2}-\frac{R_4}{R_3+R_4}=\frac{R_1\cdot R_3-R_2\cdot R_4}{(R_1+R_2)(R_3+R_4)}$$

Note 1-1: For the sake of clarity, it is more convenient to consider the ratio of the output and the input voltage U_A / U_E. Therefore, the following equations are set up for this ratio.

Book: Hoffmann, Karl. *Applying the Wheatstone bridge circuit*. Germany: HBM, 1974.

اندازه گیری بوسیله پل وتستون



$$\frac{U_A}{U_E} = \frac{R_1}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} = \frac{R_1 \cdot R_3 - R_2 \cdot R_4}{(R_1 + R_2)(R_3 + R_4)}$$

balanced bridge

$$\frac{U_A}{U_E} = 0$$

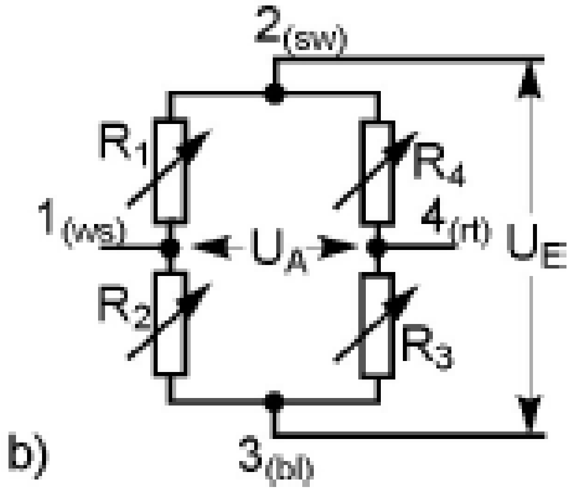
if $R1 = R_2 = R_3 = R_4$

or $R_1 : R_2 = R_4 : R_3$

Note 1-2: In all practical cases where strain gages are used, the balanced state is only reached with more or less deviation. Therefore, all instruments designed for strain gage measurements are equipped with balancing elements, which allow the indication to be set to zero for the initial state. This allows the use of equation (3) for all further considerations.

Another assumption is that the bridge output remains unloaded electrically. The internal resistance of the instrument connected to this output must be large enough to prevent noticeable errors. This is assumed for all further considerations.

اندازه گیری بوسیله پل وتستون

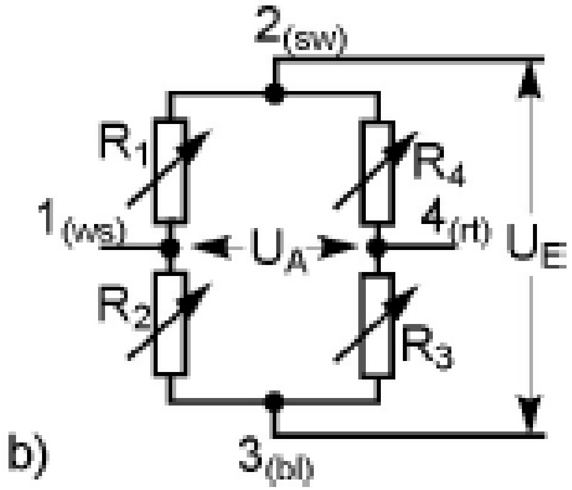


$$\frac{U_A}{U_E} = \frac{R_1}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \equiv \frac{R_1 \cdot R_3 - R_2 \cdot R_4}{(R_1 + R_2)(R_3 + R_4)}$$

If the resistors R_1 to R_4 vary, the bridge will be detuned and an output voltage U_A will appear. With the additional assumption that the resistance variation ΔR_i is much smaller than the resistance R_i itself (which is always true for metal strain gages), second order factors can be disregarded. We now have the following relationship

$$\frac{U_A}{U_E} = \frac{1}{4}\left(\frac{\Delta R_1}{R_1} - \frac{\Delta R_2}{R_2} + \frac{\Delta R_3}{R_3} - \frac{\Delta R_4}{R_4}\right) \qquad (4)$$

Book: Hoffmann, Karl. *Applying the Wheatstone bridge circuit*. Germany: HBM, 1974.

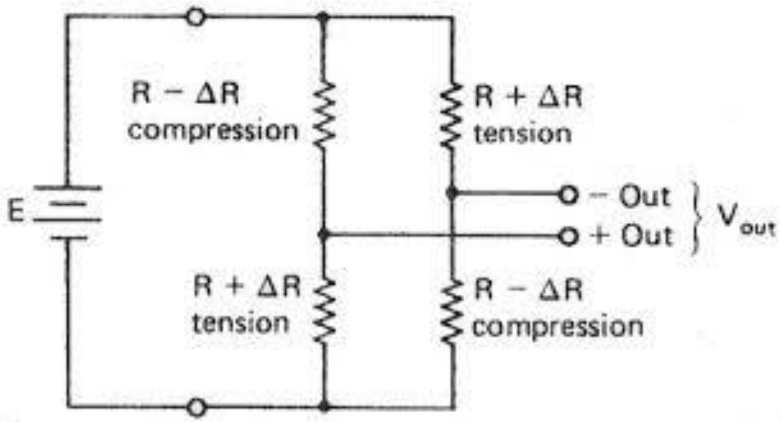


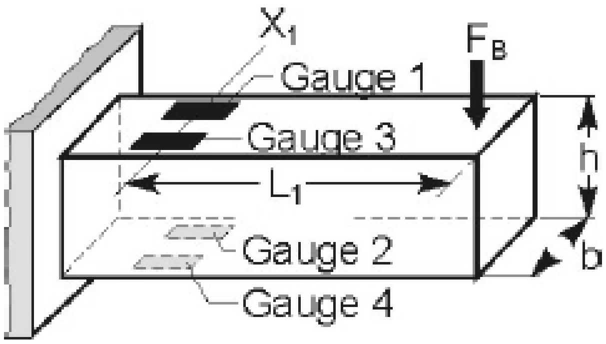
$$\frac{U_A}{U_E} = \frac{1}{4}\left(\frac{\Delta R_1}{R_1} - \frac{\Delta R_2}{R_2} + \frac{\Delta R_3}{R_3} - \frac{\Delta R_4}{R_4}\right)$$

Note 1-3: For practical strain gage applications, the pairs R_1, R_2 and R_3, R_4 or all four resistors R_1 to R_4 should have the same nominal value to ensure that the relative changes of the individual bridge arms are proportional to the relative variation of the output voltage. It is not significant whether R_1 and R_4 (or R_2 and R_3) have the same or a different nominal resistance. This is the reason for always assuming $R_1 = R_2 = R_3 = R_4$.

Substituting equation (1) ($\Delta R/R_0 = k \cdot \varepsilon$) in (4) we find

$$\frac{U_A}{U_E} = \frac{k}{4} \cdot (\varepsilon_1 - \varepsilon_2 + \varepsilon_3 - \varepsilon_4)$$

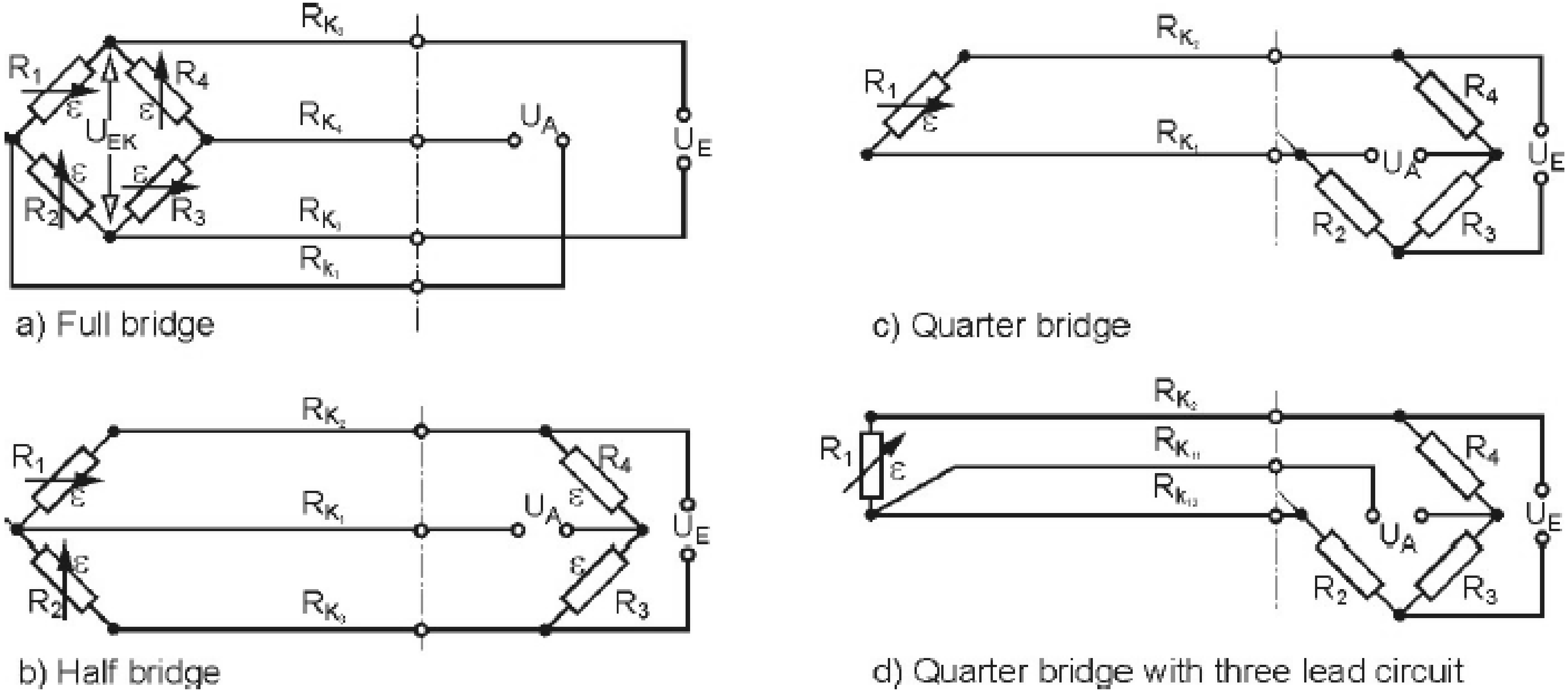




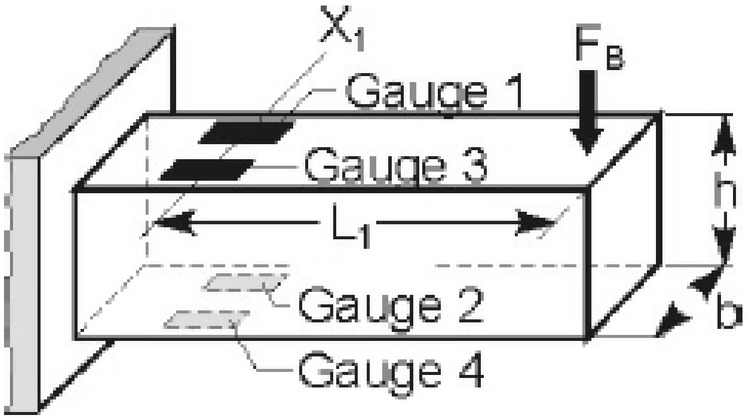
$$V_{out} = \frac{R + \Delta R}{R + \Delta R + R - \Delta R} \times E - \frac{R - \Delta R}{R - \Delta R + R + \Delta R} \times E = \frac{\Delta R}{R} \times E$$

$$\frac{V_{out}}{E} = \frac{\Delta R}{R} = k \times \varepsilon$$

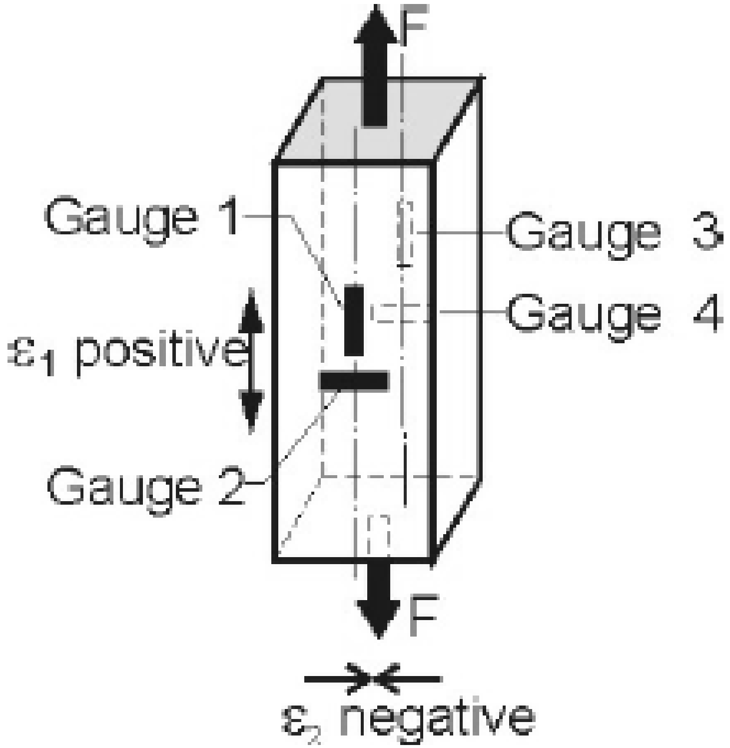
اثر مقاومت سیمها در اندازه گیری به روش پل:



Measurements on a bending beam



Measurements on a tension bar



کاربردهای از اندازه گیری بوسیله پل وتستون

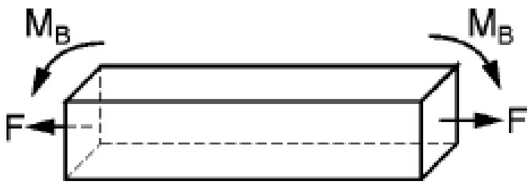


Fig. 7: Tension bar with bending moment superimposed

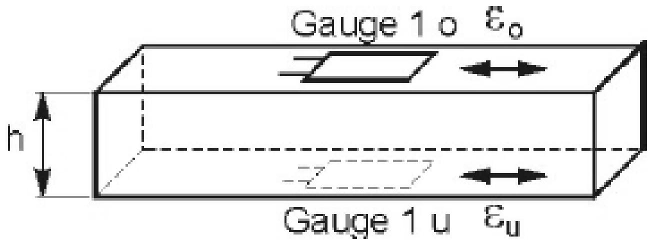


Fig. 8: Tension/bending beam according to fig. 7 with strain gages applied


سوال؟